



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



Nicolaus Copernicus
Astronomical Center
of the Polish Academy of Sciences

Radiation Physics for SMBH Accretion Disks in PLUTO

CAMK PAN SUMMER STUDENT PROGRAM

Fritz Ali Agildere

September 11, 2026

Completed Work



Overview

Background

- Introduction

- Definitions

Method

- Implementation

- Results

- Outlook

Overview

Background

Introduction

Definitions

Method

Implementation

Results

Outlook

Overview

Background

Introduction

Definitions

Method

Implementation

Results

Outlook

Overview

Background

Introduction

Definitions

Method

Implementation

Results

Outlook

Overview

Background

Introduction

Definitions

Method

Implementation

Results

Outlook

Overview

Background

Introduction

Definitions

Method

Implementation

Results

Outlook

Thank You!

- A. Mignone et al., *PLUTO: A Numerical Code for Computational Astrophysics* (ApJ) 2007
- M. Čemeljić, “Atlas” of Numerical Solutions for Star-Disk Magnetospheric Interaction (A&A) 2019
- M. Čemeljić, W. Kluźniak, S. Kundu, *Magnetic Precession as a Model for Quasiperiodic Oscillations in Pulsating Ultra Luminous X-Ray and Flat-Top Noise Sources* (A&A) 2025

Resources on GitHub: [here](#)